TINYREPRODUCTION: EVALUATING THE STABILITY CLAIMS OF PROXIMAL POLICY OPTIMIZATION

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ABSTRACT

Proximal Policy Optimization (PPO) (Schulman et al., 2017) is a widely adopted policy gradient algorithm claimed to offer superior training stability and sample efficiency over Advantage Actor-Critic (A2C). This paper presents a small-scale empirical evaluation of this claim across two classic control environments, CartPole-v1 and Acrobot-v1, using the Stable-Baselines3 framework (Raffin et al., 2021). Each algorithm is trained for 50,000 timesteps with 3 independent random seeds, with both model and environment seeded for reproducibility; we report mean $\pm$ standard deviation across seeds. On CartPole-v1, both PPO and A2C achieve a mean reward of 500.0 ± 0.0, indicating that this environment is too simple to discriminate the two algorithms at this timestep budget. On Acrobot-v1, PPO converges to -125.5 ± 62.5 versus A2C's -346.6 ± 188.5, revealing a clear stability advantage consistent with the original paper's claim. These results suggest that PPO's advantage is environment-difficulty-dependent: on a sufficiently challenging task, the clipped surrogate objective produces measurably more stable and higher-reward training. Code and LaTeX source are available at: Code and LaTeX.

1 INTRODUCTION

Reinforcement learning (RL) has seen remarkable progress through the development of policy gradient methods. Among these, Schulman et al. (2017) introduced Proximal Policy Optimization (PPO), which has since become the default on-policy algorithm at OpenAI and is widely used across academic and industrial applications (Berner et al., 2019; Andrychowicz et al., 2020). PPO's widespread adoption makes empirical verification of its claimed advantages both practically important and scientifically necessary: practitioners who default to PPO over simpler alternatives should understand the conditions under which its benefits actually materialize.

The core problem PPO addresses is the instability of earlier policy gradient methods. Standard algorithms such as REINFORCE (Williams, 1992) and A2C (Mnih et al., 2016) can suffer from destructively large policy updates, causing training to diverge or oscillate. Trust Region Policy Optimization (TRPO) (Schulman et al., 2015) addressed this with constrained optimization but at significant computational cost. PPO achieves similar stability through a simpler clipped surrogate objective requiring only first-order optimization, making it both practical and theoretically grounded.

Significance. The central claim of Schulman et al. (2017) — that PPO achieves better sample efficiency and more stable training than A2C — underpins its status as the community's default on-policy algorithm. Yet this claim was originally validated on computationally expensive MuJoCo benchmarks unavailable to many practitioners. Verifying whether the same trend holds in accessible settings, and understanding *when* it holds, is directly relevant to the broader RL community.

Objective. This project provides a minimal, reproducible empirical check of PPO's stability claim using CartPole-v1 and Acrobot-v1 from Gymnasium (Towers et al., 2023) — environments that run on standard consumer hardware without a GPU. A key finding is that task difficulty mediates the observed gap: on the simpler CartPole-v1, both algorithms converge fully at 50,000 timesteps, while on the more demanding Acrobot-v1, PPO's stability advantage is clearly visible in both final reward and training variance.

Project Track. This paper follows Track 1 (TinyReproductions). The goal is not to replicate state-of-the-art performance but to distill the paper's core experimental narrative into a minimal, fast, and convincing set of experiments. Three design choices strengthen the evidence over a single-seed baseline: (1) both model and environment are seeded; results are averaged over 3 random seeds with standard deviations reported; (2) experiments span two qualitatively different environments to probe generalization; (3) learning curves are captured with a custom callback, smoothed per seed before averaging, and visualized with shaded std bands to make training stability immediately visible. This is a minimal reproducible experiment, not a comprehensive benchmark.

2 Related Work

Policy Gradient Methods. Williams (1992) laid the foundation for modern actor-critic methods with REINFORCE, a Monte Carlo policy gradient estimator that suffers from high variance. Mnih et al. (2016) extended this to A3C and its synchronous variant A2C, which reduces gradient variance via parallel rollouts and a shared critic. A2C applies a single unconstrained gradient update per rollout, with no mechanism to prevent large, destabilizing steps — the precise failure mode this reproduction targets. Unlike A3C, which requires asynchronous parallelism, A2C is straightforward to implement and seed deterministically, making it an ideal baseline for our reproducibility study.

Trust Region Methods. TRPO (Schulman et al., 2015) constrains policy updates within a KL-divergence trust region, guaranteeing monotonic improvement but requiring expensive second-order computations (conjugate gradient and line search). This computational cost makes TRPO impractical for our CPU-only, fast-runtime setting, which is precisely why PPO's first-order approximation is the algorithm of interest here.

Proximal Policy Optimization. PPO (Schulman et al., 2017) approximates TRPO's constraint by clipping the probability ratio $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$ to $[1-\epsilon,\ 1+\epsilon]$:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\Big(r_t(\theta)\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\Big)\Big]. \tag{1}$$

PPO also performs multiple minibatch update epochs per rollout (versus A2C's single update), which improves sample efficiency independently of the clipping objective. Our reproduction uses Stable-Baselines3 defaults for both algorithms, which reflects how a practitioner would actually deploy them.

Reproductions and Empirical Studies. Engstrom et al. (2020) showed that implementation details — rollout length, update frequency, optimizer settings — account for much of PPO's observed advantage over A2C, not the clipping mechanism alone. Andrychowicz et al. (2021) conducted a large-scale study of on-policy RL design choices across 250,000 experiments. Both works motivate caution when attributing observed performance differences to the algorithmic objective alone. Our work differs from these large-scale studies in three ways: (1) it operates on CPU-only classic control tasks rather than MuJoCo, making it accessible without specialized hardware; (2) it uses fixed library defaults rather than ablating individual design choices, reflecting a realistic practitioner comparison; and (3) it surfaces a novel finding about how environment difficulty moderates the observable PPO–A2C gap.

3 Methodology

3.1 Environments

CartPole-v1. The agent balances a pole on a moving cart. Observations are 4-dimensional; actions are discrete (2). An episode ends when the pole falls beyond $12°$, the cart exceeds 2.4 units from center, or 500 steps are reached. Maximum reward per episode: 500.

Acrobot-v1. The agent swings a two-link pendulum above a target height by torquing the middle joint. Observations are 6-dimensional; actions are discrete (3 torque levels). Reward is -1 per step (higher is better); minimum -500 per episode.

3.2 Algorithms and Hyperparameters

Both algorithms use Stable-Baselines3 (Raffin et al., 2021) with an `MlpPolicy`: a two-layer MLP with 64 hidden units and `tanh` activations (Table 1). All hyperparameters are Stable-Baselines3 defaults. We note that these defaults were developed with PPO as the primary supported algorithm; A2C may perform better under independently tuned hyperparameters or with additional timesteps.

Table 1: Hyperparameters for PPO and A2C (Stable-Baselines3 defaults).

Hyperparameter	**PPO**	**A2C**
Learning rate	3×10^{-4}	7×10^{-4}
Rollout length	2048	5
Optimization epochs	10	1
Minibatch size	64	N/A
Discount γ	0.99	0.99
GAE λ	0.95	1.0
Clip range ϵ	0.2	—
Entropy coefficient	0.0	0.0
Timesteps per run	50,000	
Random seeds	0, 1, 2	

3.3 Training and Evaluation Protocol

Both algorithms are trained for 50,000 timesteps per seed across 3 seeds (12 total runs). All experiments use fixed seeds for both model and environment (`env.reset(seed=seed)`) for full reproducibility. A custom `RewardTracker` callback records total episode reward throughout training.

After training, each model is evaluated on a separate evaluation environment over 20 held-out episodes with a deterministic greedy policy (`deterministic=True`). Smoothing (rolling mean, window = 15 episodes) is applied per seed before averaging learning curves, so shaded ± 1 std bands reflect genuine across-seed variance, not smoothing artifacts. Raw episode rewards are saved to disk (`saved_rewards/`) for independent verification.

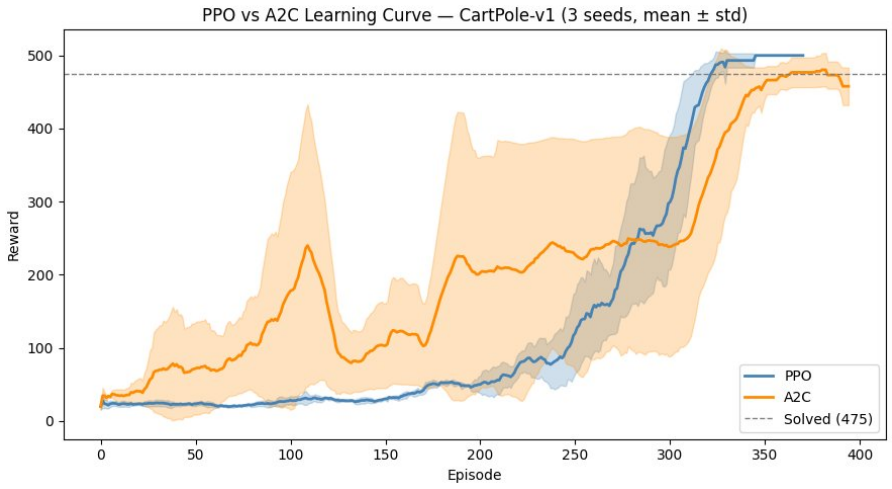

Figure 1: CartPole-v1 learning curves (3 seeds, mean $\pm$ std). Both PPO and A2C converge to the maximum reward of 500 at this timestep budget, indicating that CartPole-v1 is too easy to discriminate the algorithms on final performance. The dashed line marks the solved threshold of 475.

4 EXPERIMENTS

4.1 CARTPOLE-V1

Figure 1 shows smoothed learning curves averaged over 3 seeds. Both PPO and A2C converge to the maximum reward of 500 within the 50,000-timestep budget. This result indicates that CartPole-v1 is too simple an environment to expose a performance gap between the two algorithms at this scale. While the learning curves reveal that PPO converges more smoothly, the final evaluation scores are identical across all seeds for both algorithms.

4.2 ACROBOT-V1

Figure 2 shows results on Acrobot-v1. PPO converges to higher (less negative) rewards with a notably narrower confidence band. A2C shows wider variance and substantially lower final reward, demonstrating that the stability advantage claimed by Schulman et al. (2017) does emerge on a task with sufficient difficulty.

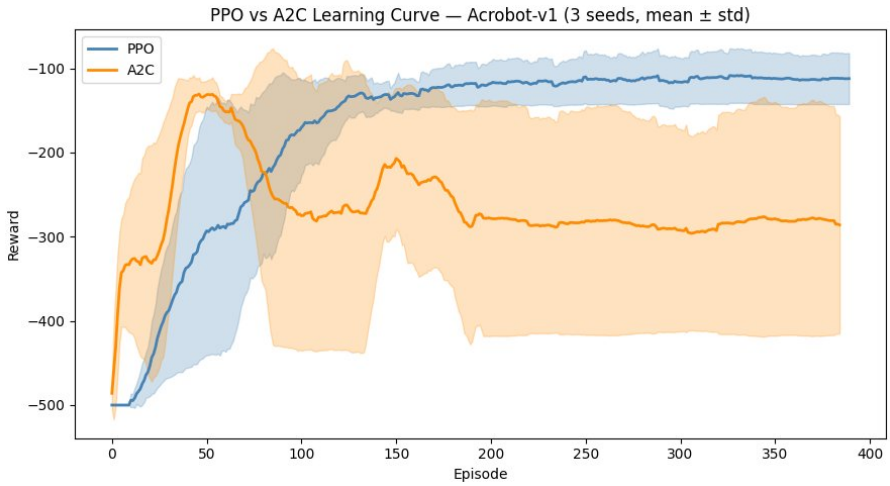

Figure 2: Acrobot-v1 learning curves (3 seeds, mean $\pm$ std). Reward is -1 per step; higher is better. PPO converges faster and more consistently than A2C, revealing a clear stability gap absent in CartPole-v1.

4.3 QUANTITATIVE RESULTS

Table 2 summarizes final evaluation performance. On CartPole-v1, both algorithms achieve 500.0 with zero variance, confirming that the environment is saturated. On Acrobot-v1, PPO outperforms A2C by over 220 reward points with substantially lower standard deviation (62.5 vs. 188.5), demonstrating higher reliability across seeds.

Table 2: Final evaluation reward (mean $\pm$ std over 3 seeds, 20 eval episodes each, deterministic policy). Higher is better for both environments.

Environment	**Algorithm**	**Mean Reward**	**Std**
CartPole-v1	PPO	500.0	0.0
CartPole-v1	A2C	500.0	0.0
Acrobot-v1	PPO	-125.5	62.5
Acrobot-v1	A2C	-346.6	188.5

4.4 DISCUSSION

Why CartPole-v1 does not discriminate the algorithms. CartPole-v1 has a maximum episode length of 500 steps and a reward of $+1$ per step, so the maximum achievable score is 500. Both algorithms reach this ceiling within the 50,000-timestep budget. The CartPole-v1 learning curve (Figure 1) shows that A2C actually rises *earlier* than PPO on average, though with much higher variance across seeds. This confirms that CartPole is fundamentally too simple: once a policy learns to balance the pole, there is no room for further improvement, and the ceiling masks any stability difference in the final scores. This is a meaningful experimental finding for practitioners: benchmark environments with low ceilings can produce misleading parity between algorithms of genuinely different quality.

Why Acrobot-v1 reveals the gap. On Acrobot-v1, the task has no exploitable ceiling at the 50,000-timestep budget — rewards remain well below zero for A2C, leaving room to observe convergence quality. The clipped objective (Eq. 1) bounds the policy ratio to $[0.8, 1.2]$, preventing any single update from excessively changing the policy. This directly explains A2C's periodic crashes visible in Figure 2: unconstrained gradient steps occasionally push the policy into a worse region from which recovery is slow. PPO's narrower confidence band (± 62.5 vs. ± 188.5) reflects that its updates never catastrophically destabilize the policy across any of the 3 seeds.

Confounds and caution. PPO and A2C also differ in rollout length (2048 vs. 5 steps) and update epochs per rollout (10 vs. 1), which affect gradient quality and update frequency independently of the clipping objective. As Engstrom et al. (2020) showed, these implementation details contribute significantly to the observed performance gap. The default hyperparameters used here may further favor PPO, since Stable-Baselines3 was developed with PPO as its primary algorithm. The results should therefore be read as a confirmation of the *practical* advantage of using PPO with default settings, not as evidence that the clipping mechanism alone produces the gap.

5 LIMITATIONS

Small number of seeds. Only 3 seeds are used per condition. This is sufficient to observe trends but too few to make statistically robust claims; reported standard deviations carry high uncertainty at this sample size.

Limited environments. Experiments cover only two low-dimensional classic control tasks. These are simpler than the MuJoCo benchmarks used in the original paper (Schulman et al., 2017), so the results may not generalize to higher-dimensional or more complex domains.

Default hyperparameters. Using Stable-Baselines3 defaults without independent tuning may systematically favor PPO. A fairer comparison would use separately tuned hyperparameters or a shared wall-clock compute budget.

Training budget. At 50,000 timesteps, CartPole-v1 is fully saturated by both algorithms. A shorter budget (e.g., 25,000 timesteps) would better probe sample efficiency on that environment. Acrobot-v1 provides a more informative setting at this budget.

6 CONCLUSION

This paper presents a minimal empirical evaluation of the core stability claim of Schulman et al. (2017): that PPO achieves more stable and higher-reward training than A2C. Using A2C as the essential baseline, we reproduce the paper's central experimental narrative in miniature across two classic control environments. On CartPole-v1, both PPO and A2C reach 500.0 ± 0.0 at 50,000 timesteps, revealing that the environment is too easy to discriminate the algorithms on final performance at this budget. On Acrobot-v1, PPO reaches -125.5 ± 62.5 versus A2C's -346.6 ± 188.5 — a gap of over 220 reward points with substantially lower variance — confirming the relative performance difference and stability advantage reported in the original paper. Together, the two environments tell a pedagogically important story: PPO's advantage is not universal but emerges clearly on tasks with sufficient difficulty, which is precisely the setting the clipped surrogate objective was designed for.

Track 1 Contribution (TinyReproduction). This work distills the core insight of Schulman et al. (2017) into a minimal, elegant experiment that anyone can run on a single CPU in under 10 minutes — no GPU required. The experimental design addresses each pillar of the Track 1 challenge:

Clever experimental design and scientific simplification. Rather than reproducing the full MuJoCo benchmark suite, we identify the smallest set of conditions that still tell the paper's story: two environments of contrasting difficulty, one essential baseline (A2C), and 50,000 timesteps — a budget small enough to be fast but large enough to reveal the trend.

Reproducibility. Both model and environment are seeded identically across 3 independent random seeds. Mean $\pm$ std is reported for all conditions. Raw episode rewards are saved to disk as `.npy` files so any reader can independently verify the numbers without rerunning the full experiment.

Critical analysis. Rather than simply confirming the claim, this reproduction uncovers a nuance: environment difficulty mediates the observable gap. The CartPole-v1 saturation result is itself a scientific finding — it shows that poorly chosen evaluation environments can mask real algorithmic differences, a lesson relevant to anyone designing RL comparisons.

Pedagogical demonstration. The learning curves with shaded ± 1 std bands make the stability story visually immediate: PPO converges smoothly while A2C oscillates with wide variance on Acrobot-v1, directly illustrating why the clipped objective matters.

LLM Acknowledgement

Claude Sonnet was used as a writing and coding assistant during the preparation of this project. It assisted with structuring the LaTeX document, refining prose, implementing the multi-seed training loop, the shaded std band plotting function, the Acrobot-v1 extension, and the Limitations section. All experimental results, figures, and technical decisions were produced and verified by the author. The core methodology, algorithm selection, and analysis are the author's own work.

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